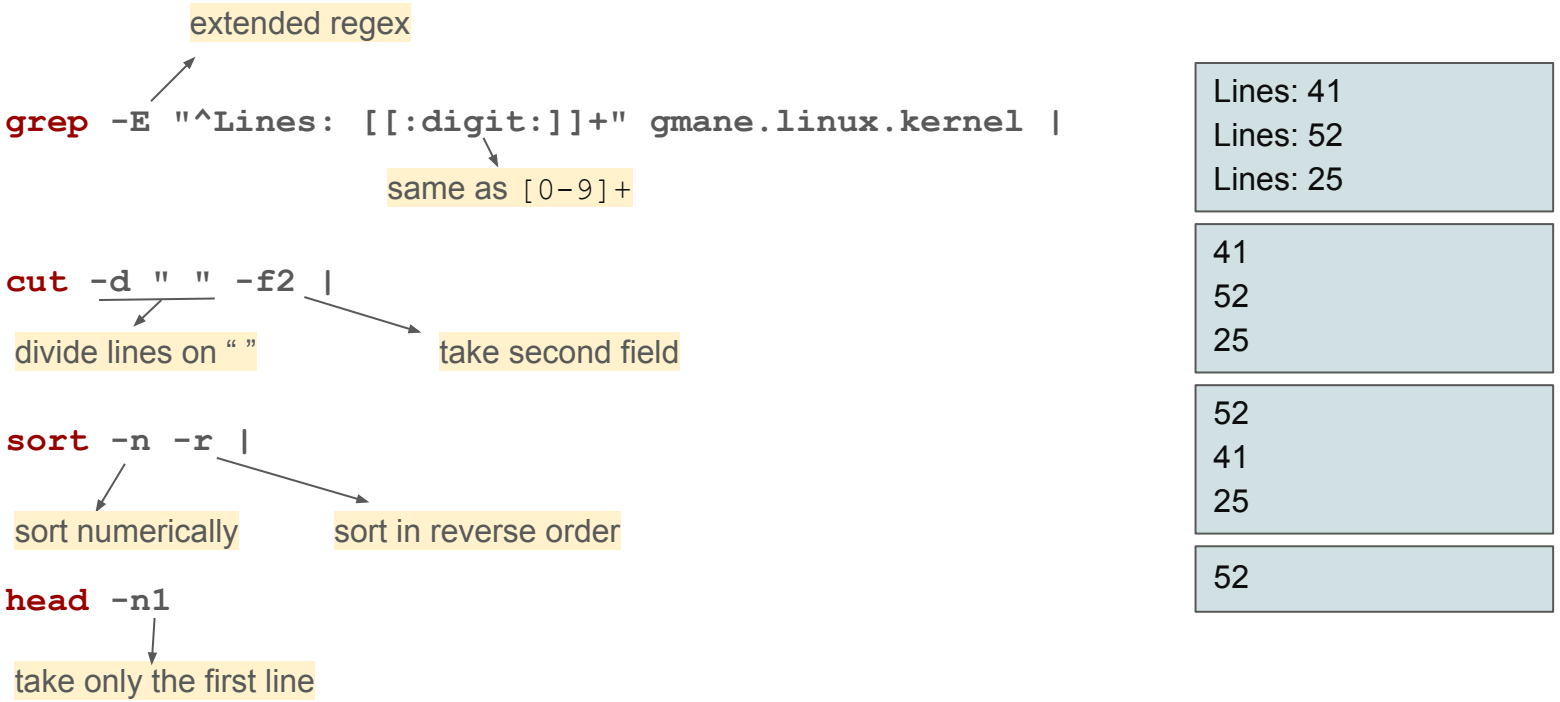
Solutions

FDE Tutorial Sessions, Week 2

Task 5.1


extended regex
grep -E "^Lines: [[:digit:]]+" gmane.linux.kernel |
same as [0-9]+
cut -d " " -f2 |
divide lines on " "
take second field
sort -n -r |
sort numerically
sort in reverse order
head -n1
take only the first line
Lines: 41
Lines: 52
Lines: 25
41
52
25
52
41
25
52

Task 5.2

```
grep "^In-Reply-To:" gmane.linux.kernel |
```

start of line

only the reply header lines

```
cut -d " " -f2-1000 |
```

divide on “ ”

fields 2 … 1000 = addresses (assume at most 999)

```
sort |
```

equal addresses end up next to each other

```
uniq -c |
```

count each run of equal lines (needs `sort` first!)

```
sort -n -r |
```

most replies first (numeric + reverse order)

```
head -n1
```

the mail with the most replies

```
In-Reply-To: <a@mail>
In-Reply-To: <b@mail>
In-Reply-To: <a@mail>
```

```
<a@mail>
<b@mail>
<a@mail>
```

```
<a@mail>
<a@mail>
<b@mail>
```

```
2 <a@mail>
1 <b@mail>
```

```
2 <a@mail>
1 <b@mail>
```

```
2 <a@mail>
```

Task 5.3

```
grep "^Lines: " gmane.linux.kernel |
awk '{s += $2; n++} END {print s/n}'
```

every “Lines:” header = one per mail

add field 2 (the number) to s

count mails

runs for every input line

runs once, after the last line

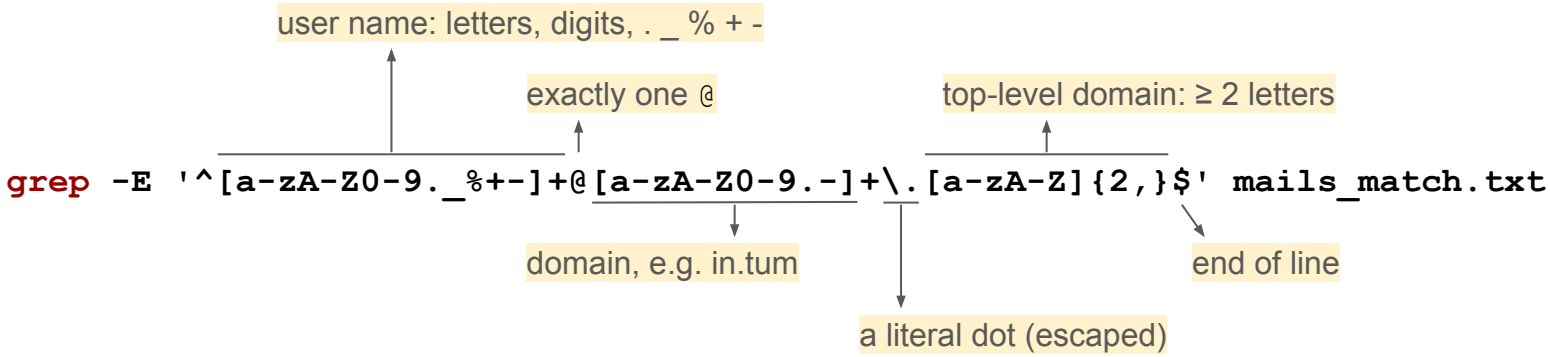
sum / count = average

Lines: 41
Lines: 52
Lines: 25

s = 41 + 52 + 25 = 118
n = 3

118 / 3 = 39.33

Task 6.1



john.doe@in.tum.de → match	john.doe@tum → no match (no .tld)

Task 6.2

print only the matched part

```
grep -E -o '@.*' mails_match.txt |
```

from @ to end of line = the domain

```
sort |
```

equal domains next to each other

```
uniq -c |
```

count each domain

```
sort -n -r
```

most frequent domain first

```
ann@vger.kernel.org
bob@in.tum.de
eve@vger.kernel.org
```

```
@vger.kernel.org
@in.tum.de
@vger.kernel.org
```

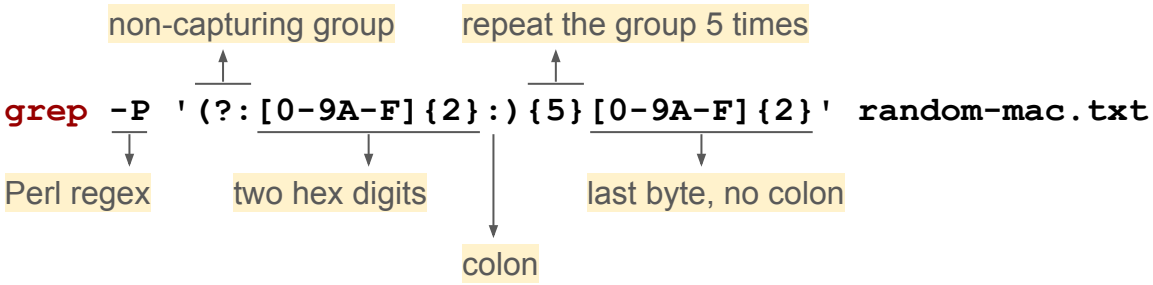
```
@in.tum.de
@vger.kernel.org
@vger.kernel.org
```

```
1 @in.tum.de
2 @vger.kernel.org
```

```
2 @vger.kernel.org
1 @in.tum.de
```

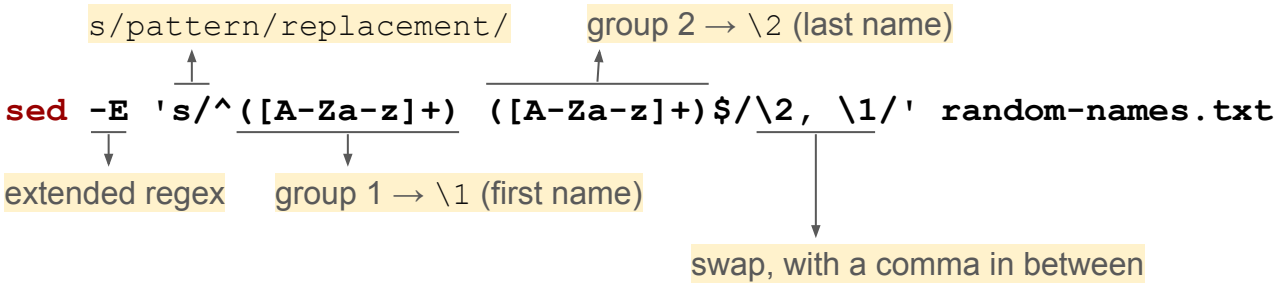
Task 7.1

`-P` is needed because `grep -E` does not have `(?:…)`




```
(3A:)(F1:)(07:)(BC:)(9E:) 24
  └──── group × 5 ────┘  last
```

Task 7.2



John Doe

→ \1 = John, \2 = Doe →

Doe, John